

Machine Learning with Python

Lecturer: Dr. I. U. Hewapathirana

Data Preprocessing

Terms to Describe Data

variables

samples

number	air_pressure_9am	air_temp_9am	avg_wind_direction_9am
0	918.060000	74.822000	271.100000
1	917.347688	71.403843	101.935179
2	923.040000	60.638000	51.000000
3	920.502751	70.138895	198.832133
4	921.160000	44.294000	277.800000

Alternative Names

- Sample
 - ✓ Record
 - ✓ Example
 - ✓ Row
 - ✓ Instance
 - ✓ Observation
- Variable
 - ✓ Feature
 - ✓ Column
 - ✓ Dimension
 - ✓ Attribute
 - ✓ Field

Data Types

Numerical (Quantitative)

- Height
- Score on an exam
- Number of transactions per hour
- Change in stock price

Categorical (Qualitative/Nominal)

- Color of an item
- Gender
- Marital Status
- Type of customer
- Product categories

Other

- String
- Date

Data Types

Numerical (Quantitative)

- Values are numbers
- Quantitative
- Height
- Score on an exam
- Number of transactions per hour
- Change in stock price

Categorical (Qualitative/Nominal)

- Values are labels, names or categories
- Qualitative or nominal
- Color of an item
- Gender
- Marital Status
- Type of customer
- Product categories

Other

- String
- Date

Data Types

Numerical (Quantitative)

- Height
- Score on an exam
- Number of transactions per hour
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Categorical (Qualitative/Nominal)

- Color of an item
- Gender
- Marital Status
- Type of customer
- Product categories

Other

- String
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Exploratory Data Analysis

- Summary statistics – information that summarizes a dataset
- Data visualization – look at data graphically

Example Summary Statistics

Statistic	Value
Mean	3.76
Median	4.35
Standard deviation	1.77
Minimum	1.00
Maximum	6.90

Summary statistics help us understand the centre, spread and range of a variable.

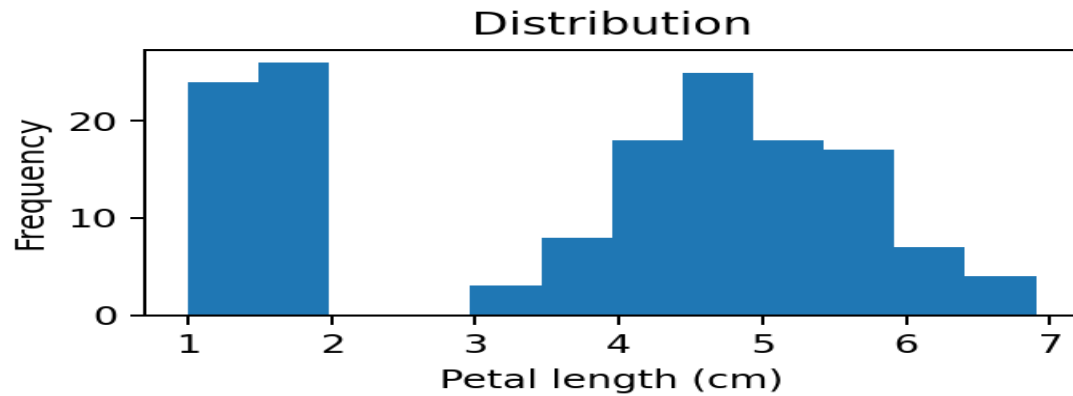
Types Of Data Visualizations

- **Distribution**: how a variable or variables in the dataset distribute over a range of possible values.
- **Relationship**: how the values of multiple variables in the dataset relate
- **Composition**: how the dataset breaks down into subgroups
- **Comparison**: how trends in multiple variable or datasets compare
- For more details see <https://python-graph-gallery.com/>

Example Data Visualizations

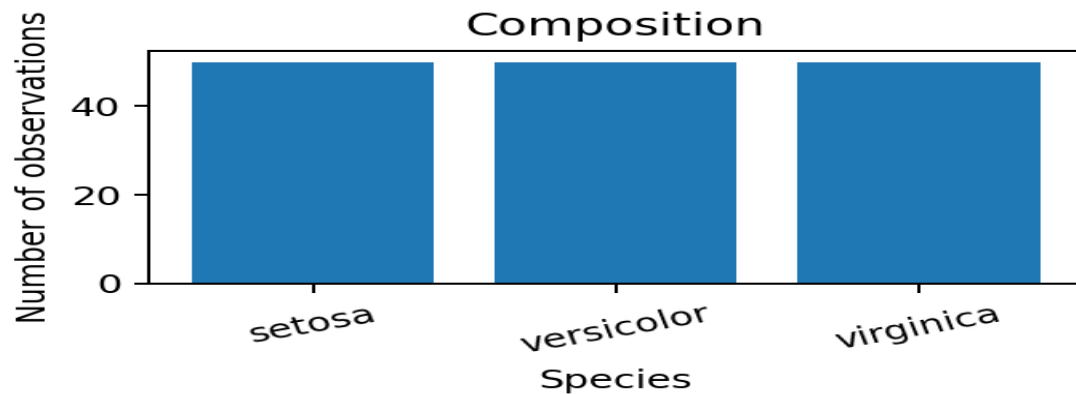
Python libraries: pandas + matplotlib

Distribution



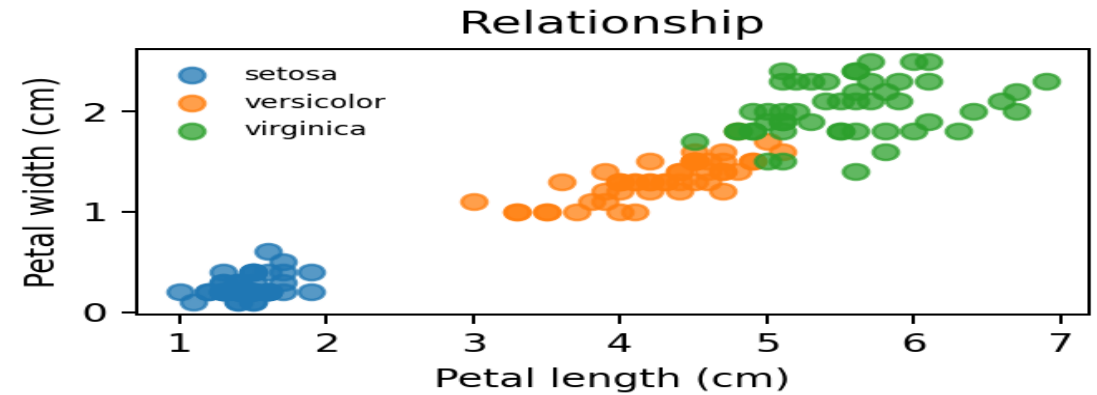
```
plt.hist(df['petal_length'])
```

Composition



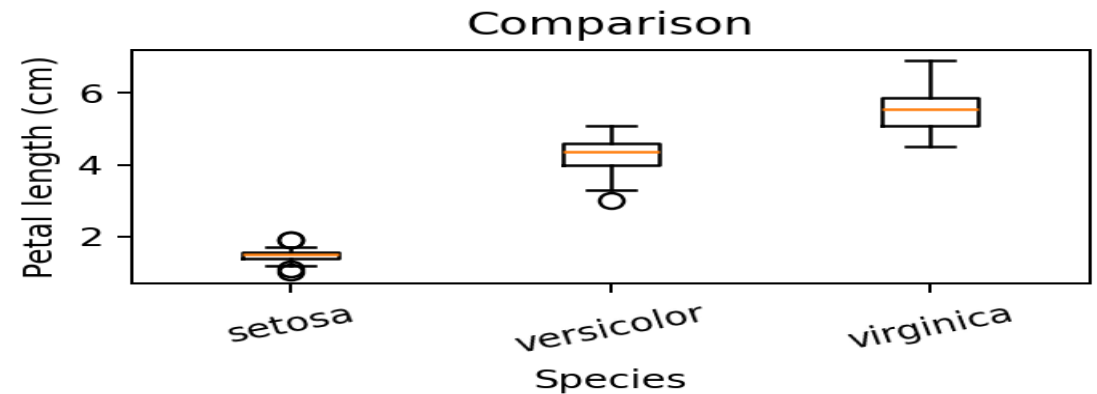
```
df['species'].value_counts().plot(kind='bar')
```

Relationship



```
plt.scatter(df['petal_length'], df['petal_width'])
```

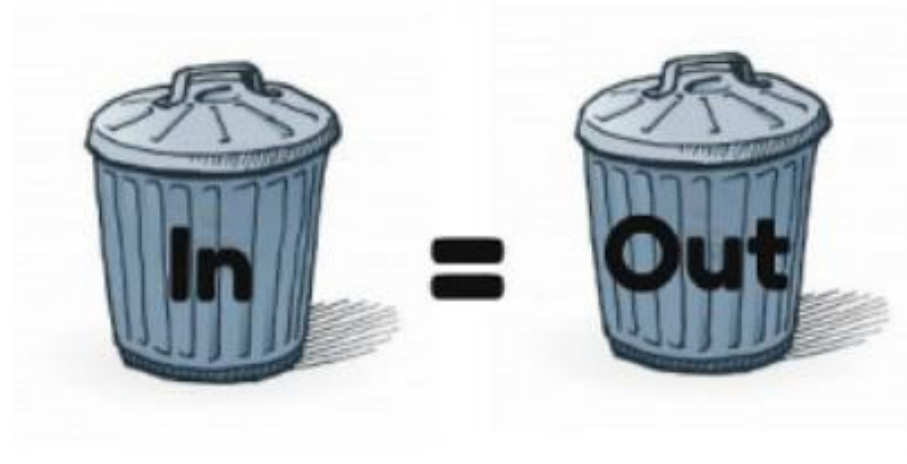
Comparison



```
plt.boxplot(groups)
```

Preparing Data for Analysis

- Addressing data quality issues – Data Cleaning
- Getting data in shape – Data Munging/Data Wrangling/ Data Preprocessing
 - Feature selection
 - Feature transformation



Data Quality Issues

- Missing Data
- Duplicate Data
- Invalid Data
- Noise
- Outliers

Missing Data

Name	Age	Income
Angela	34	80
Sidney	--	56
Ratan	10	--
Kiril	68	--
Zhou	45	120

Missing values could be: NaN, empty string, ?, -1, -99, -999 and so on.

Addressing Missing Data - Remove

Name	Age	Income
Angela	34	80
Sidney	--	56
Ratan	10	--
Kiril	68	--
Zhou	45	120



Name	Age	Income
Angela	34	80
<i>Sidney</i>	<i>--</i>	<i>56</i>
<i>Ratan</i>	<i>10</i>	<i>--</i>
<i>Kiril</i>	<i>68</i>	<i>--</i>
Zhou	45	120

Addressing Missing Data - Impute

Name	Age	Income
Angela	34	80
Sidney	--	56
Ratan	10	--
Kiril	68	--
Zhou	45	120



Name	Age	Income
Angela	34	80
Sidney	50	56
Ratan	10	50
Kiril	68	50
Zhou	45	120

Replace missing values with something reasonable

- Mean
- Median
- Most frequent
- Sensible value based on application

Duplicate Data

Name	Address
Angela	430 Park Drive
Sidney	7800 West View Street
Sid	7800 West View Street
Ratan	12442 Mountain Avenue
Kiril	45 East 5 th St
Kiril	1220 Mill Avenue
Zhou	4345 Apple Lane

Addressing Duplicate Data – Delete/Merge

- Delete older record.
- Merge duplicate records

Name	Address
Sidney	7800 West View Street
Sid	7800 West View Street
Kiril	45 East 5 th St
Kiril	1220 Mill Avenue



Name	Address
Sidney	7800 West View Street
Sid	7800 West View Street
Kiril	45 East 5th St
Kiril	1220 Mill Avenue

Invalid Data

Name	Zip Code
Angela	346412
Sidney	92618
Ratan	8033A
Kiril	11012
Zhou	59285

Addressing Invalid Data

- Use external data source to get correct value
- Apply reasoning and domain knowledge to come up with reasonable value.

Name	Zip Code
Angela	346412
Ratan	8033A



Name	Zip Code
Angela	34641 2
Ratan	8033 A

Domain Knowledge

- Required for addressing data quality issues effectively
 - How the data is collected
 - The user population
 - Intended user application

Feature Selection

- Goal: Select smallest set of features that best captures data for application
- Domain knowledge is important aka 'feature engineering'
- Methods:
 - Remove feature
 - Features that are very correlated
 - Features with a lot of missing values
 - Irrelevant features: ID, row number, etc.
 - Re-code feature
 - Discretization: re-format continuous feature as discrete
 - New features derived from existing features
 - Combine features
 - E.g.: Height , Weight → BMI
- Feature Selection in Python with Scikit-Learn will be covered later during the course

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Feature Selection

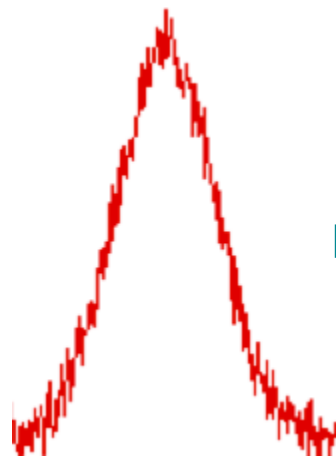
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Feature Selection

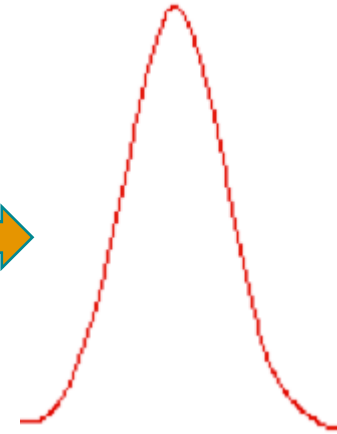
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 - Re-code feature
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 - New features derived from existing features
 - **Combine features**
 - E.g.: Height , Weight → BMI
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Feature Transformation

- Map feature values to new set of values



Original Data



Transformed Data

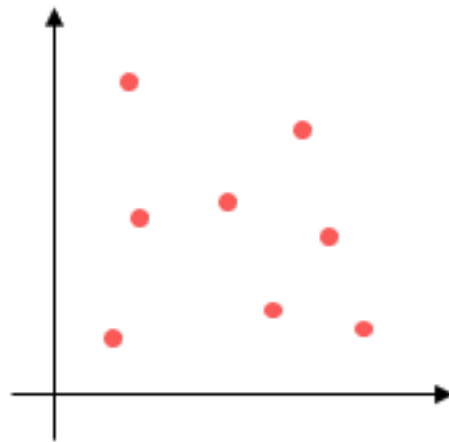
- Have data in a format suitable for analysis
- Take care not to filter out important characteristics of data

Feature Transformation - Scaling

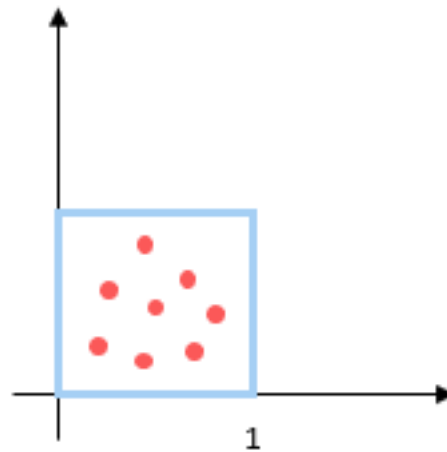
- Changing the range of values of a feature or features to another specified range
- Avoid some features that dominate the analysis

Feature	Value 1	Value 2	Value 3
Age (years)	25	45	65
Income (\$)	30,000	60,000	120,000
Number of Purchases	3	6	12

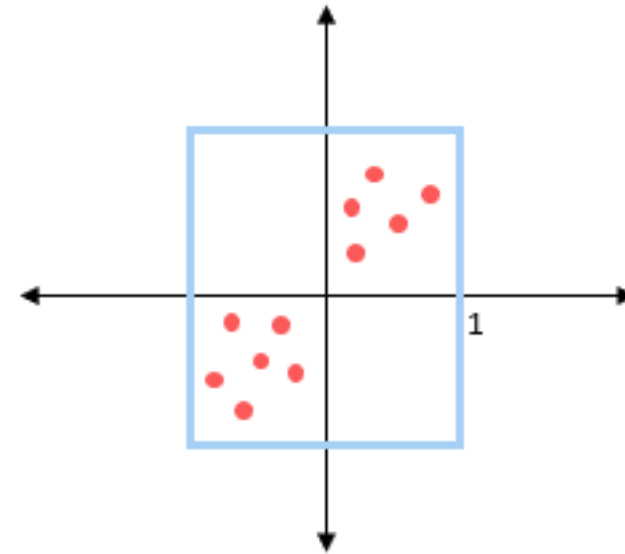
Scaling



Actual Data

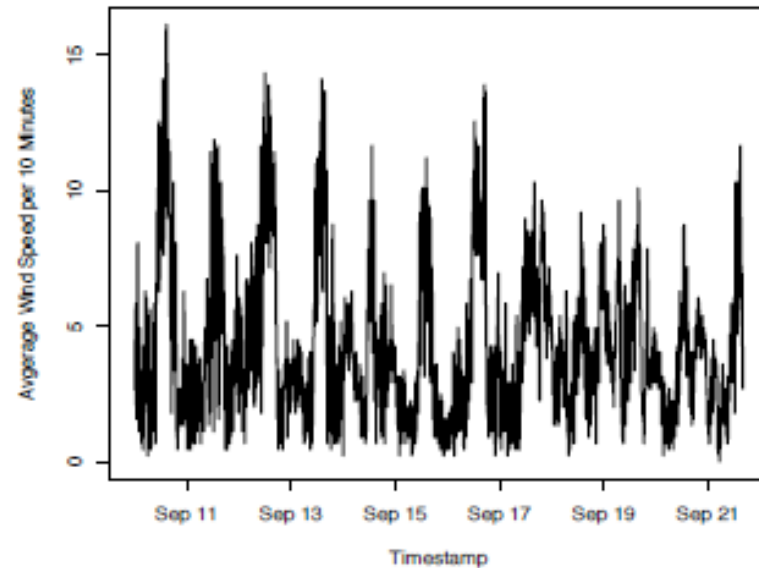


After normalizing

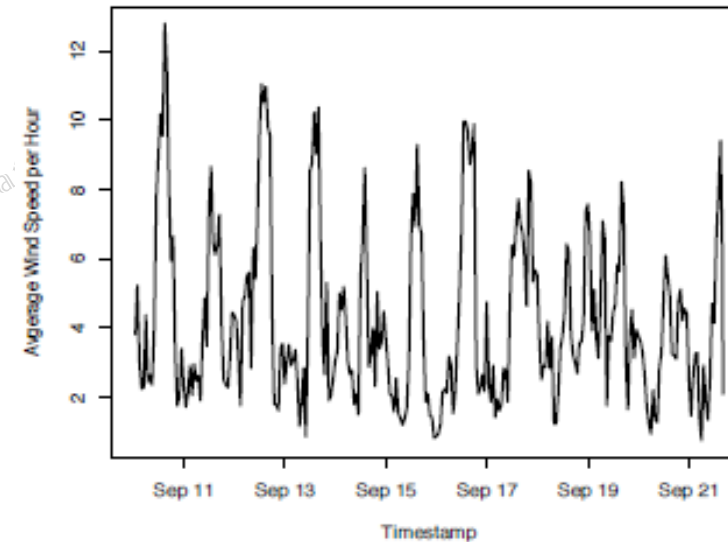


After standardization

Feature Transformation - Aggregation



Avg Wind Speed
(every 10 minutes)



Avg Wind Speed
(every 60 minutes)